

Ionic and Covalent Bonds

Answer key with worked solutions

A. Understanding

1. Ionic vs covalent.

Ionic: electrons are transferred from one atom (usually a metal) to another (usually a non-metal), and the oppositely charged ions attract. **Covalent:** two atoms (usually non-metals) share one or more pairs of electrons.

2. Electronegativity.

How strongly an atom attracts the shared pair of electrons in a covalent bond.

B. Classifying bonds

3. Classify.

(a) Cl-Cl: 0.00, **nonpolar covalent** (b) C-Cl: $3.16 - 2.55 = 0.61$, **polar covalent** (c) N-H: $3.04 - 2.20 = 0.84$, **polar covalent** (d) Mg-O: $3.44 - 1.31 = 2.13$, **ionic** (e) Li-F: $3.98 - 0.98 = 3.00$, **ionic**

4. Partial negative charge.

C-Cl: **Cl** is δ^- . N-H: **N** is δ^- .

5. Increasing polarity.

ΔEN : C-H 0.35, H-Br 0.76, O-H 1.24. Order: **C-H (least polar), H-Br, O-H (most polar)**

C. Ionic formulas

6. Formulas.

(a) K^+ and O^{2-} : K_2O (b) Ca^{2+} and Br^- : CaBr_2 (c) Al^{3+} and N^{3-} : AlN (d) Mg^{2+} and P^{3-} : Mg_3P_2

7. Ion formation.

$\text{Mg} \longrightarrow \text{Mg}^{2+} + 2\text{e}^{-}$ and $\text{Cl}_2 + 2\text{e}^{-} \longrightarrow 2\text{Cl}^{-}$. Overall the compound is MgCl_2 .

D. Properties

8. Melting points.

NaCl is an ionic lattice: melting requires overcoming many **strong attractions between oppositely charged ions**, which takes a lot of energy. Methane is made of separate molecules: melting only overcomes the **weak attractions between molecules**. The covalent C-H bonds do not break.

E. Challenge

9. Conductivity.

Conduction needs **charged particles that can move**. In the solid, the ions are held in fixed positions in the lattice. When molten or dissolved, the ions are free to move, so they can carry charge.